

Order parameter and mechanism of the RBLapSum through_rank_kappa instability

Research note: one-step score-scale dynamics

2026-09-16/17. Follow-up to the kappa stability investigation.

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1 Purpose and summary of conclusions

The previous investigation (docs/rblapsum-kappa-stability-neutral.tex) introduced the geometric indicator

$$\chi_{\text{geo}} = \frac{b}{2T\delta}$$

and used it as an empirical instability proxy. This note treats that proxy as a hypothesis and asks four questions: whether χ_{geo} is the right control parameter (H1); whether the dimensionless surrogate gain $\Pi = \|L_{\kappa}D_z\|_F^2$ is a better geometric quantity (H2); what dynamical mechanism produces score-inflation bursts (H3); and whether $n_{\text{eff}} \rightarrow 1$ is a cause or a late-stage consequence (H4).

The measured answers, stated fully in the final sections:

- **H1 (proxy): supported.** χ_{geo} is not reparameterization-invariant; an exact score-units transformation leaves the training dynamics identical while scaling χ_{geo} arbitrarily, and within a fixed (K, T) row χ_{geo} has no prospective signal (AUC ≈ 0.46).
- **H2 (surrogate gain): supported as the gain measure.** $\Pi = \|L_\kappa D_z\|_F^2$ is dimensionless, invariant, predicts the realized gradient gain (rank correlation 0.97 vs 0.52 for χ_{geo}), and its dense-boundary approximation $b^2/(4T\delta)$ holds within a factor of two in healthy states.
- **H3 (coupling to scale): supported in the diffusion/heating form.** The signed surrogate drift is restoring at the dangerous states; the destabilizing coupling is the second-order heating of the scale by large parameter steps (even under sign flip of the gradient, $\propto \text{lr}^2$, and at hot states deterministic rather than minibatch noise). Instability occurs where heating exceeds the restoring drift.
- **H4 ($n_{\text{eff}} \rightarrow 1$ late-stage): supported with a refinement.** The one-feature limit does null the corrected gradient (measured $\Pi = 0$), but near-degenerate blocks continue to pump; post-collapse growth stops only when the support gradient is disabled, and kernel broadening after the collapse makes things worse.
- **Practical controller variables:** per-block window population (n_{eff} , the best within-row early warning, AUC 0.75) for the fast guard; Π as the portable pressure gauge; support-gradient shutdown, not kernel broadening, as the correct emergency action after a collapse.

2 Definitions

All quantities are computed per layer from the candidate set of the gate (the top $K+J$ features by score).

- Scores $s_i = |z_i|$; boundary $b = \max(b_0, s_{(K+1)})$; kernel $\kappa_i = e^{-|s_i-b|/T}/(2T)$; $Z = \sum_i \kappa_i$; $n_{\text{eff}} = Z^2/\sum_i \kappa_i^2$; local spacing $\delta = (s_{(K-3)} - s_{(K+5)})/8$.
- The kappa-corrected support gradient, written with $D_\kappa = \text{diag}(\kappa)$ and $L_\kappa = D_\kappa - \kappa\kappa^\top/Z$, is

$$g_s = -L_\kappa w, \quad w_i = -u_i z_i, \quad u_i = \partial L / \partial y_i,$$

equivalently $g_s = L_\kappa D_z u$ up to the sign convention.

- Exact surrogate gain $\Pi = \|L_\kappa D_z\|_F^2$, computed per token from the diagonal-minus-rank-one structure:

$$\Pi = \sum_j z_j^2 \kappa_j^2 \left[(1 - \kappa_j/Z)^2 + (\sum_i \kappa_i^2 - \kappa_j^2)/Z^2 \right].$$

Π is dimensionless (each factor $\kappa_j z_j$ is a pure number), unlike χ_{geo} , which carries units of 1/score. The dense-boundary approximation tested below is $\Pi \approx b^2/(4T\delta)$.

- Empirical gain $G_{\text{emp}} = \|g_s\|^2/\|u\|^2$ over candidates.
- Scale statistics on a fixed evaluation batch: $R_{\text{all}} = \text{RMS}_i(s_i)$ over all features, $R_K = \text{mean of the top-}K \text{ scores}$, and b ; the scale variable is $x = \log R$.
- One-step counterfactual updates from a saved model+optimizer state: branches *null* (zero gradient), *task* (support gradient disabled), *full*, and their antithetic negatives. The saved Adam-type state replays momentum on every branch, and any parameter perturbation inflates R at second order (heating), so the signed drift is the antithetic odd part $\mu = \frac{1}{2} \mathbb{E}[\Delta x(+G) -$

$\Delta x(-G)]$ and the heating is the even part relative to the null step. The support contribution is $\text{odd}(\text{full}) - \text{odd}(\text{task})$; raw-gradient decomposition $G_{\text{full}} = G_{\text{task}} + G_{\text{supp}}$ is exact (verified; identical RNG, no stochastic layers), while optimizer steps are nonlinear in the gradient and are always reported as separate branch counterfactuals.

3 Experiments performed

1. **Checkpoint library.** Thirteen states spanning: calm ($T=2$), stable ($k32/T1$), a marginal survivor around a recoverable burst ($k32\ j224\ T0.5$ at steps 2000/4000/6000), marginal states of runs that later died ($k32\ j480\ T0.5$; $k64\ j448\ T1$), and a post-collapse state ($k32\ j480\ T0.1$). All checkpoints carry full optimizer state.
2. **One-step drift/diffusion** (16 batches per state, five branch steps per batch, per-layer $\Delta \log R$ on a fixed evaluation batch).
3. **Surrogate gain and radial structure** per state and layer: Π , $b^2/(4T\delta)$, χ_{geo} , n_{eff} , G_{emp} , the radial projection $r_{\text{supp}} = -s^\top g_s^{(z)}/s^\top s$, $(s-b)^\top L_\kappa w$, $\text{Cov}_q(s-b, w)$, and a permutation null (48 shuffles of w among candidates per token).
4. **Scale-response curves** $F(\alpha)$: a gate-level function-preserving rescaling $z \rightarrow \alpha z$ (output divided by α) leaves the forward and the task gradient exactly unchanged (unit-tested and verified in the measurements) while the surrogate sees an α -times score excursion at fixed T ; drift branches are measured at $\alpha \in \{0.5, 0.75, 1, 1.25, 1.5, 2, 3\}$ at four reference states.
5. **Score-units reparameterization:** $\alpha = c$ together with $T \rightarrow cT$ must leave all gradients invariant while $\chi_{\text{geo}} \rightarrow \chi_{\text{geo}}/c$.
6. **Drift vs. heating scaling:** learning-rate multipliers $\{0.5, 1, 2\}$ and half batch size at three states.
7. **Counterfactual continuations** from a marginal $k96/T1$ state and from the post-collapse $T=0.1$ state: normal replicas (three seeds), support gradient disabled from the start or at a boundary-inflation trigger, kernel broadened ($T \times 3$, or $T \times 30$ post-collapse) from the start or at the trigger.
8. **Prospective order-parameter comparison** on 148 checkpoint-states from 18 earlier fixed- T runs: does a quantity measured at step r predict a boundary burst in $(r, r+H]$, pooled and within (K, T) rows.

4 Results 1: the reparameterization test (H1)

The gate-level reparameterization $(s, z, b, T) \rightarrow (cs, cz, cb, cT)$ leaves the forward function and every parameter gradient invariant. This is proved exactly in unit tests (float32, single gate) and confirmed on full checkpoints: at $c = 0.5$ and $c = 2$ (both exact in floating point) the training loss is unchanged to all printed digits and the worst relative parameter-gradient difference is 2×10^{-3} – 6×10^{-3} , attributable to bf16 accumulation order. At $c = 3$ (inexact mantissa scaling) near-tied scores reorder in bf16 and the candidate sets change, which shifts the loss by 10^{-3} ; this is a numerics artifact, not a violation of the invariance.

Under the same transformation $\chi_{\text{geo}} \rightarrow \chi_{\text{geo}}/c$ (measured: layer-4 value 54.1 at $c=1$, 27.0 at $c=2$, 18.1 at $c=3$), while Π is invariant. Identical training dynamics with different χ_{geo} values means a universal numerical threshold in χ_{geo} cannot be fundamental.

H1 verdict: supported. χ_{geo} is an empirical proxy. It separates (K, T) rows because it co-moves with the true geometry, but it is not reparameterization-invariant, and (Results 5) it carries no within-row temporal signal.

5 Results 2: the surrogate gain Π (H2)

Across 96 layer-states (12 checkpoints $\times$ 8 blocks, excluding the post-collapse state), the exact Π predicts the realized gradient gain G_{emp} with rank correlation 0.97; χ_{geo} reaches 0.52. The dense-boundary approximation $\Pi \approx b^2/(4T\delta)$ holds within a factor of about two at the mid-depth blocks of healthy states and fails, as expected, when the boundary region degenerates (factor 10 during late inflation; factor 10^4 post-collapse). It can also *overestimate* by orders of magnitude at individual blocks whose local rank spacing collapses (near-tied scores make $\delta \rightarrow 0$ while the exact Π stays moderate) — visible as the rightmost points of Fig. 1; where it matters, the exact Π should be used.

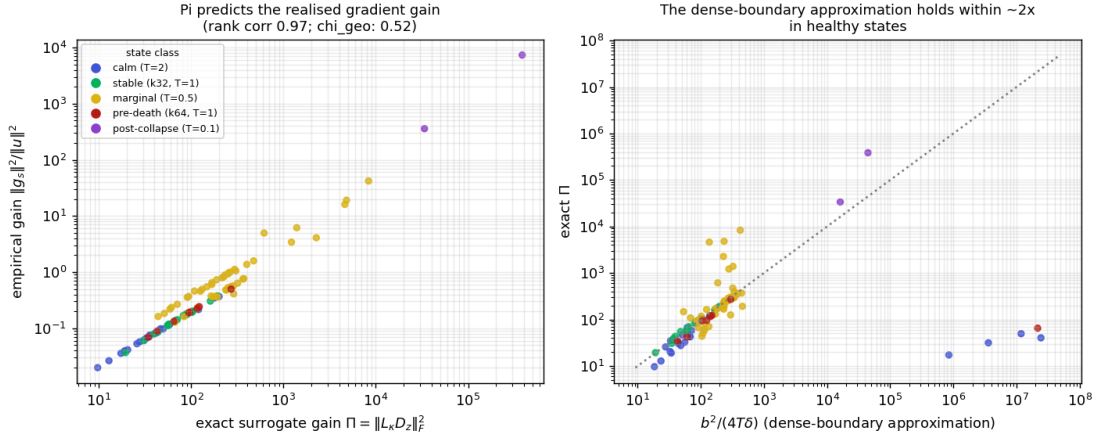


Figure 1: Left: exact Π against the realized gain $\|g_s\|^2/\|u\|^2$ across states and blocks. Right: exact Π against the dense-boundary approximation. Each point is one (checkpoint, block) pair; color encodes the checkpoint’s state class: calm ($T=2$), stable ($k32/T1$), marginal ($T=0.5$), pre-death ($k64/T1$), post-collapse ($T=0.1$).

The predicted K - and T -ratios from the earlier report are reproduced by the approximation ($\Pi_{\text{approx}}(k64, T1)/\Pi_{\text{approx}}(k32, T1) = 1.66$ at step 2000, in the predicted 1.6–2.25 range) but compressed in the exact value (ratio 1.09), because the exact Π weights the true kernel and z profile rather than a uniform dense boundary.

H2 verdict: supported as a gain measure. Π is dimensionless, reparameterization-invariant, and is the correct predictor of how much gradient the surrogate injects. Whether large Π implies instability is a separate question (H3).

6 Results 3: signed drift versus heating (H3)

Three facts emerge.

1. **The signed surrogate drift is not the runaway force.** At the state 1,940 steps before the $k64/T1$ death, the surrogate’s odd part is *strongly restoring* (-5.7×10^{-2} per step pooled; -0.10 at the worst block), and the task odd part is also restoring (-2.2×10^{-2}). The same holds during late inflation of the dying $T=0.5$ run.

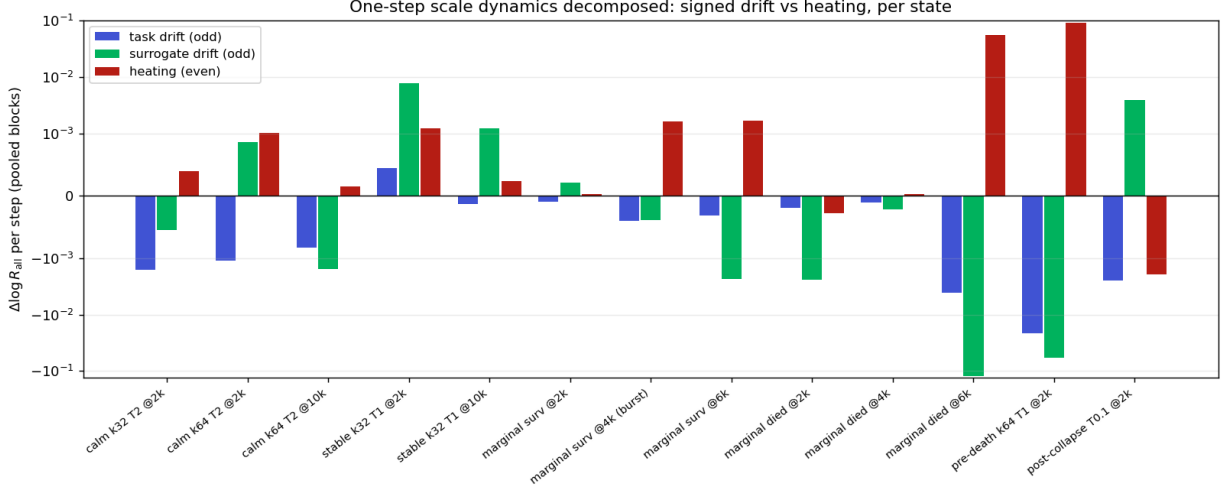


Figure 2: One-step change of $\log R_{\text{all}}$ per training step, decomposed into signed task drift, signed surrogate drift (antithetic odd parts) and heating (even part), pooled over blocks, at thirteen states.

2. **The dominant destabilizing term is heating:** the even, sign-independent inflation of R produced by the sheer size of the parameter step ($\mathbb{E}[(z+\delta z)^2] > z^2$). At the pre-death state it reaches $+9.2 \times 10^{-2}$ per step pooled and $+0.18$ at the worst block — larger than the restoring drift. At calm and stable states heating is 30–300 times smaller. The learning-rate sweep confirms the identification: the odd parts scale approximately linearly with the step size, the even part approximately quadratically (Fig. 3).
3. **The heating at the pre-death state is deterministic, not minibatch noise:** halving the batch size leaves it unchanged ($+9.3 \times 10^{-2}$ vs $+9.2 \times 10^{-2}$) and its across-batch variance is small. It is second-order overshoot of a large, structured gradient, not gradient-variance diffusion. At the stable $k32/T1$ state, by contrast, the (much smaller) heating roughly doubles under batch halving, i.e. there it *is* mostly stochastic.

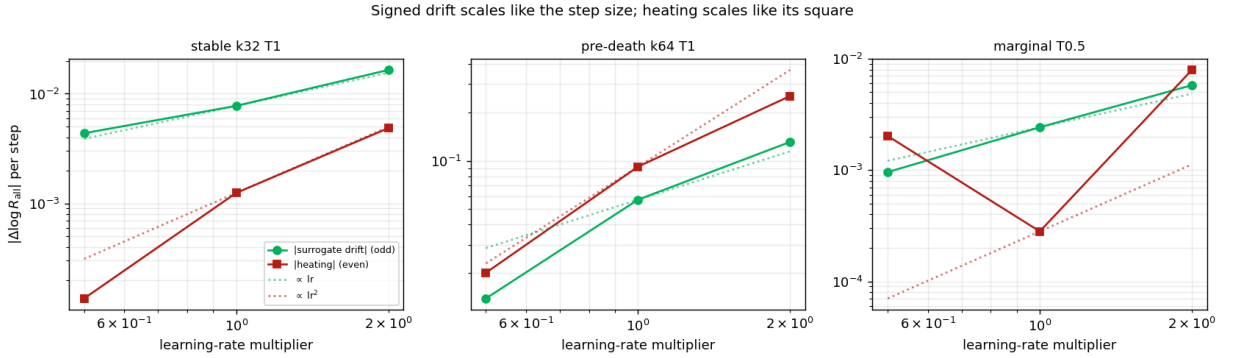


Figure 3: Drift and heating versus learning-rate multiplier at three states, with reference slopes $\propto \text{lr}$ and $\propto \text{lr}^2$.

The scale-response curves (Fig. 4) characterize the signed part: the surrogate behaves as a spring in the gate-view scale. At the calm $k64/T2$ state it crosses zero near $\alpha \approx 1.4$: it pushes the scale up below that point and down above it — a stable fixed point. At the stable $k32/T1$ state the spring pushes up over the whole measured range with decreasing strength. The task drift is exactly α -independent by construction of the intervention, and measured to be so, which also validates the isolation of the surrogate path.

The permutation test on the rank structure of w shows that the signed radial component of

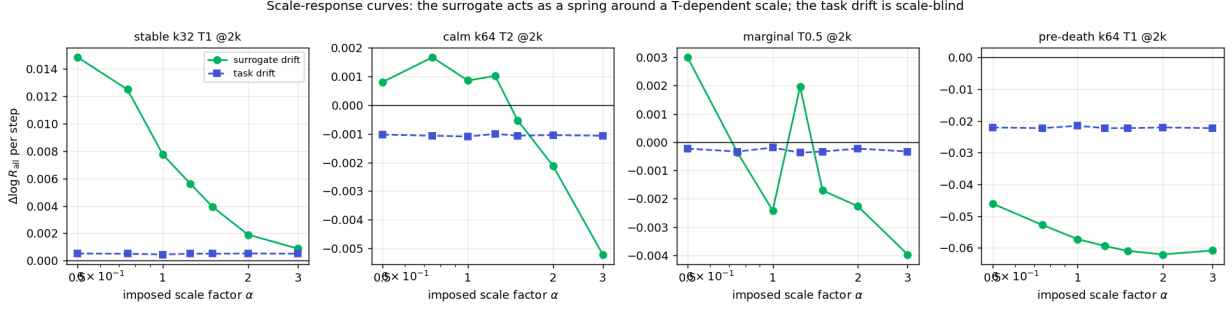


Figure 4: Surrogate and task signed drift as functions of an imposed function-preserving scale factor α at four states.

the surrogate is statistically distinguishable from a rank-shuffled null in most states (typically $|z\text{-score}| \gtrsim 3\text{--}20$), with an outward-aligned sign at the pre-death state ($+19.8\sigma$) and inward-aligned signs at calm late states. The magnitudes are, however, small compared with the heating term at the dangerous states, so the sign structure sets the (mostly restoring) spring, while the danger comes from the even term.

H3 verdict: supported in its second form. Large surrogate gain is not sufficient for instability through a signed radial force; the coupling to scale growth is primarily through second-order heating — deterministic step-size overshoot at the hot states — opposed by a restoring drift. Instability corresponds to heating exceeding restoration, layer by layer.

7 Results 4: counterfactual interventions and the death arc (H4, causality)

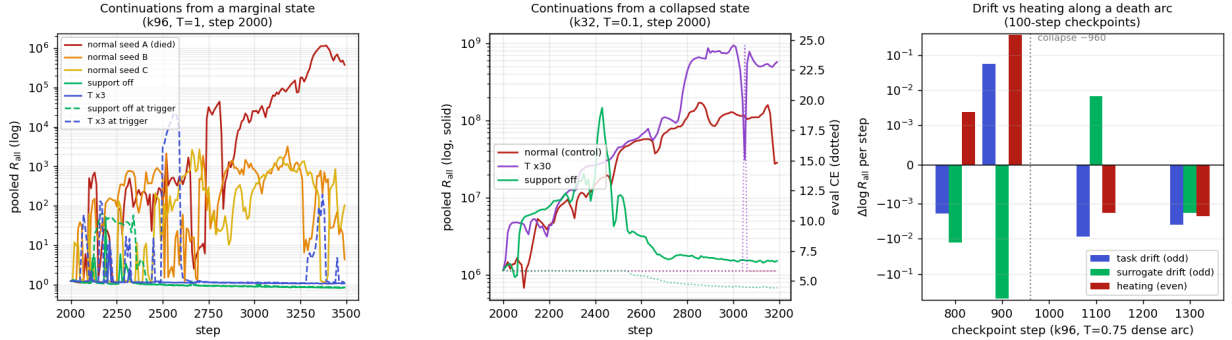


Figure 5: Left: continuations from a marginal $k96/T1$ state (step 2000); three normal seeds against support-off and $T \times 3$, applied from the start (solid) or at a boundary-inflation trigger (dashed). Middle: continuations from a collapsed $T=0.1$ state; solid lines show scale, dotted lines evaluation CE. Right: the one-step decomposition along a dense death arc ($k96, T=0.75$, checkpoints every 100 steps; collapse at approximately step 960).

From the marginal state ($k96/T1$, 1500-step continuations): all three normal replicas show repeated scale bursts (excursions of one to three orders of magnitude that mostly recover); one of three diverges terminally. With the support gradient disabled, the trajectory is entirely calm and the scale *decreases* (task gradient alone is restoring, consistent with the drift measurements); with $T \times 3$ the trajectory is calm at unchanged scale and reaches the best evaluation CE of all branches (1.505 against 1.552–1.634). Both remedies also work when applied only at the first trigger (boundary above three times its rolling median). The support surrogate is therefore *necessary* for burst initiation at this state.

From the collapsed state ($T=0.1$, scale $\sim 10^6$): the control continues to inflate ($\times 25$ in 1200 steps) with CE frozen near 5.8. Broadening the kernel ($T \times 30$) *accelerates* the inflation ($\times 500$): it re-populates the kernel window at a state where some blocks have astronomically large Π (up to 10^{12} ; one block sits exactly at $n_{\text{eff}} = 1$ with $\Pi = 0$, the measured $L_\kappa \rightarrow 0$ limit). Disabling the support gradient *stops* the inflation and the model begins to recover (CE $5.93 \rightarrow 4.44$ in 1200 steps). Post-collapse scale growth is thus driven by the degenerate surrogate, not by the task gradient.

Along the dense death arc (checkpoints every 100 steps): at 200 steps before collapse the decomposition is ordinary (restoring drift -1.3×10^{-2} , heating $+2.4 \times 10^{-3}$, already ten times the calm level); at 60–100 steps before collapse both terms explode two orders of magnitude (surrogate drift -0.50 , heating $+0.37$ per step, pooled) — a violent tug-of-war whose balance tips block by block; after collapse all terms are small again (kernel largely dead).

H4 verdict: supported with a refinement. $n_{\text{eff}} \rightarrow 1$ is a late-stage phenomenon, and the strict one-feature limit does eliminate the corrected support gradient ($\Pi = 0$, measured). The refinement is that the near-degenerate blocks that remain ($n_{\text{eff}} \approx 2$, very large z) continue to pump scale: the late runaway is still surrogate-driven, in its degenerate form. “Loss of the stabilizing distributed correction” and “continuing surrogate engine” are therefore both partially correct: the *distributed* correction is lost, and what remains of the surrogate is destabilizing.

8 Results 5: prospective comparison of order parameters

On 148 checkpoint-states from 18 earlier fixed- T runs (post-collapse rungs excluded), predicting whether a boundary burst occurs within the next H steps:

H	pooled AUC			within-row AUC		
	χ_{geo}	Π_{approx}	$1/n_{\text{eff}}$	χ_{geo}	Π_{approx}	$1/n_{\text{eff}}$
1000	0.81	0.88	0.77	0.46	0.63	0.75
2000	0.78	0.86	0.71	0.48	0.68	0.75
4000	0.80	0.85	0.68	0.44	0.69	0.76

Pooled across configurations, Π_{approx} dominates. Within a fixed (K, T) row — the test that removes the between-row confound — χ_{geo} falls to chance, consistent with it being a row-level proxy, while the smallest per-block n_{eff} is the best temporal early-warning signal (window depopulation precedes bursts), with Π second. Only the three marginal rows contribute to the within-row comparison, so these numbers are indicative rather than definitive.

9 Results 6: drift versus diffusion, and the escape picture

The learning-rate and batch-size sweeps separate the two destabilization channels. Signed drift scales approximately linearly and heating approximately quadratically with the learning-rate multiplier at all three tested states. At the stable $k32/T1$ state the heating roughly doubles when the batch is halved (stochastic, gradient-variance origin); at the pre-death $k64/T1$ state it is unchanged (deterministic overshoot of a large structured gradient). The burst phenomenology of the earlier report — recoverable excursions with seed-dependent outcomes — is reproduced directly in the marginal continuations (three seeds: one death, two bursty survivals).

A quantitative Kramers-style collapse of burst rates onto a single escape number was not achieved with the available data: the restoring-curve slope λ can be estimated only coarsely (for the calm $k64/T2$ state, the surrogate spring crosses zero near $\alpha \approx 1.4$ with slope $\lambda \approx 7 \times 10^{-3}$ per unit $\log R$ per step), and only three (K, T) rows contain both outcomes. The qualitative structure is

nevertheless the escape picture: a scale basin created by restoring drift (task everywhere; surrogate around its T -dependent set point), an excitation source (heating) that grows with the surrogate gain and step size, and a cliff (window depopulation and correction degeneracy) beyond which restoration is lost.

10 Proposed causal sequence for a burst

Measured facts are marked (M); inference is marked (I).

1. At marginal pressure the per-step parameter updates near the gate are large: the surrogate gain Π is 3–10 times the calm value (M), and the total step produces second-order scale heating comparable to or exceeding the restoring drift at the worst block (M).
2. Heating episodes inflate the score scale of a block; the signed drifts (task and surrogate) oppose the inflation (M). Most excursions are therefore recovered (M: continuation replicas; earlier campaign trajectories).
3. During a large excursion the kernel window thins (n_{eff} falls; M in the $T=0.1$ anatomy and prospectively: low n_{eff} is the best within-row burst predictor), which concentrates the correction and raises the per-member gain further (I, consistent with the Π formula).
4. If the excursion reaches the near-degenerate regime, the distributed correction is lost ($\Pi \rightarrow 0$ at $n_{\text{eff}} = 1$, M) while neighbouring near-degenerate blocks acquire enormous gain ($\Pi \sim 10^{12}$, M); the remaining surrogate pumps scale (M: support-off stops it) and the task gradient alone cannot restore the model while the surrogate is active (M).
5. The runaway propagates through depth by the residual-replacement composition (earlier report) and terminates in a frozen high-scale state (M).

11 Recommended controller variables

- **Fast guard: per-block window population** (equivalently n_{eff}). It is the best within-row early warning (AUC 0.75), it is the variable whose collapse defines the cliff, and the existing servo’s population floor acts on exactly this quantity. The measured refinement: the guard must never *raise* T after a block has already collapsed; at that stage the correct action is to disable the support gradient (support scale to zero) until the scale statistics recover.
- **Pressure gauge:** Π (per block), or its forward-only approximation $b^2/(4T\delta)$. It is dimensionless, reparameterization-invariant, tracks the realized gain, and dominates the pooled prospective comparison. A trim that holds Π at a safe target is the natural upgrade of the current χ_{geo} -servo; within this architecture the two co-move (which is why the χ_{geo} -servo works), but Π is the quantity with a mechanistic meaning.
- χ_{geo} should be treated as a convenient row-level heuristic only.

The central question — what measurable quantity directly indicates that the score scale is about to become self-amplifying — has a two-level answer. The direct indicator is the one-step heating-to-restoration ratio at the worst block, measurable by the antithetic counterfactual probe (`analysis/scale_dynamics.py drift`); values approaching one from below precede collapse (0.02 at calm states, 0.7–1.8 at states that died within 2000 steps, ~ 2 at 100 steps before death). The cheap forward-only pair that tracks it is (Π , n_{eff}) per block: Π sets the step size the surrogate injects (hence the heating, which grows quadratically with it), and n_{eff} measures the distance to the degeneracy cliff. χ_{geo} predicts neither within a row; its apparent usefulness came from co-moving with Π across rows.

12 Remaining uncertainties and next experiments

- The heating term was identified as the destabilizing coupling, but its propagation constant — how much one block’s heating raises the next block’s scale within one step — was not measured; a block-resolved transfer measurement would close this.
- The one-step counterfactuals do not equal sustained closed-loop rates: the optimizer’s second-moment adaptation tempers repeated large steps. Short-horizon (10–50 step) branch continuations from the same states would quantify the closed-loop attenuation.
- The within-row comparison rests on three (K, T) rows; the interrupted fate-mapping campaign (16 configurations bracketing the boundary) would make the prospective table decisive.
- A Π -targeted servo variant has not been run; the recommendation above predicts it matches or improves on the χ_{geo} -servo and transfers across score reparameterizations by construction.
- The escape-rate collapse (E_{escape}) remains unresolved for lack of matched (λ, D, x_c) estimates across enough marginal configurations.